

Research Article

Advancing Loan Approval Prediction With SHAP-Guided Feature Selection and LIME-Based Model Interpretability in a Multiclassifier Context Through a Web-Based Application Development Approach

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In today's dynamic financial environment, bank loan approval systems are crucial for determining credit accessibility and maintaining economic stability. Efficient and accurate mechanisms help financial institutions minimize risks, enhance customer satisfaction, and make informed lending decisions. Traditional evaluation methods, however, often struggle with complex applicant data, underscoring the need for advanced, data-driven approaches. This study proposes an enhanced loan approval prediction framework that integrates SHAP-guided feature selection and LIME-based interpretability within a robust multi-classifier architecture. The methodology includes extensive data preprocessing, handling missing values, and encoding categorical variables, followed by SHAP to identify the most influential features. Using two Kaggle datasets, logistic regression achieved the highest performance, with 86.17% accuracy and 81% AUC on Dataset 1 and 99.06% accuracy on Dataset 2. LIME provided intuitive, visual explanations of model predictions, fostering transparency and trust. In addition, a user-friendly, real-time web application was developed for practical deployment. Overall, the study advances intelligent, interpretable, and efficient loan approval systems for modern banking.

Keywords: AUC-ROC curve; bank loan; classification; confusion matrix; machine learning

1. Introduction

The banking sector plays a vital role in a nation's economy, driving economic development and ensuring financial stability. Among the key services provided by banks, loan approval processes are particularly significant, as a substantial portion of a bank's revenue is derived from lending activities. As a result, the stability of the banking sector has far-reaching implications for the overall health of the economy. With the continuous expansion of the sector, the

volume of loan applications has surged, making it essential for banks to evaluate applicants' eligibility efficiently before granting loans [1]. This is crucial, as the primary goal of banks is to invest resources prudently [2]. In today's technologically advanced and complex financial environment, traditional manual methods of loan evaluation are increasingly inadequate. Bank managers, responsible for assessing applicants' eligibility and the associated risks, often face challenges due to the limitations of manual processes, which are prone to errors and inefficiencies. Such

shortcomings not only affect the financial stability of banks but also have broader economic consequences, particularly in cases where they contribute to an increase in non-performing assets (NPAs) [3]. By predicting potential defaulters in advance, banks can significantly reduce NPAs and protect their financial health. In addition, accurate loan approval predictions are critical for improving a bank's profitability [4]. Several factors are taken into consideration during the loan approval process, including debt-to-income ratios, savings or checking account status, occupation, job tenure, annual income, and homeownership [5]. Personal details such as identification, reasons for loan applications, and credit histories also play a critical role in assessing an applicant's creditworthiness. A thorough analysis of these factors is necessary to distinguish between defaulters and nondefaulters, particularly when credit history is a key consideration in the decision-making process [6]. However, as loan applications become increasingly complex and the demand for loans rises, developing an accurate and reliable predictive system remains a significant challenge [7]. In some cases, indicators such as employment tenure and debt-to-income ratios are the most critical factors in predicting whether a borrower will default on loan payments [8]. The complexities of applicant data in traditional loan approval processes arise from several factors, including heterogeneous data sources, missing or incomplete data, class imbalance, nonlinearity and feature interactions, and evolving creditworthiness criteria. The evolution of technology has introduced innovative approaches to address these challenges, particularly through the application of artificial intelligence (AI) and machine learning (ML). ML, a subset of AI, has revolutionized data analysis, enabling banks to uncover patterns and insights that were previously difficult to detect. This has led to more efficient and accurate loan approval processes in the banking sector [9]. ML has proven especially effective in enhancing predictive systems, with different algorithms offering varying levels of accuracy depending on the circumstances of each case. Moreover, data preprocessing significantly impacts prediction accuracy, as different techniques can lead to varying results. ML has already demonstrated its effectiveness across various sectors, including healthcare, education, agriculture, and banking [10]. By leveraging the vast amounts of data available, banks can use ML to better understand both internal and external factors. This enables more accurate and efficient processing of loan applications, reducing the time and effort required by manual evaluations. ML also enhances fraud detection, optimizes credit underwriting, improves regulatory compliance, and strengthens customer engagement [11]. Ultimately, ML models can be applied to assess the eligibility of individual loan applicants, benefiting both banks and customers by ensuring that only the most qualified applicants receive loans, while minimizing the risks associated with lending. The integration of ML into the banking sector represents a significant advancement in the loan approval process. By adopting these advanced technologies, financial institutions can make more informed decisions, reduce risks, and increase operational efficiency. This not only boosts bank profitability but also promotes

broader economic stability by fostering responsible lending practices. The adoption of ML in banking marks a shift toward a more resilient and robust financial ecosystem, delivering long-term benefits to both lenders and borrowers alike. Despite numerous studies applying ML algorithms to loan prediction, two gaps remain. First, most approaches focus solely on improving accuracy without explaining why decisions are made, limiting trust and adoption in financial institutions. Second, existing explainable-AI (XAI) studies often use Shapley Additive Explanations (SHAP) or Local Interpretable Model-Agnostic Explanations (LIME) separately rather than as complementary mechanisms within a unified training and evaluation pipeline. This work addresses these gaps by (1) employing SHAP for both feature selection and interpretability, (2) integrating LIME for instance-level explanation, and (3) performing cross-validated multiclassifier analysis with statistical and robustness validation. The proposed approach aims to balance predictive performance, transparency, and deployability, key requirements for practical banking systems.

The main objective of this research work is to develop an advanced ML-based loan approval prediction system that improves lending efficiency and risk management through the integration of SHAP-guided feature selection and LIME-based model interpretability, facilitating data-driven decision-making. Unlike prior studies, our work introduces the first fully integrated pipeline where SHAP is used both as a global feature selection mechanism and as an interpretability tool, while LIME provides instance-level explanations. The combined use of SHAP + LIME within a multiclassifier framework, complemented with cross-validation, significance testing, and robustness analysis, forms the core novelty of this study.

The significant contributions of this research work are summarized as follows:

1. Integration of SHAP-based feature selection for loan approval prediction.
 - Unlike conventional feature selection methods, this study employs SHAP to identify the most influential features for loan approval prediction.
 - This approach ensures that the model focuses on the most critical predictors, improving accuracy and interpretability.
2. Integration of XAI using LIME.
 - To enhance transparency in decision-making, LIME is incorporated, providing stakeholders with clear insights into individual loan approval predictions.
 - This addresses a critical gap in previous studies that prioritize accuracy over explainability.
3. Comprehensive multiclassifier evaluation on two datasets.
 - This research evaluates eight ML classifiers, including random forest (RF), AdaBoost (AdaB), XGBoost (XGB), gradient boosting (GB), Naïve Bayes (NB), support vector machine (SVM), decision tree (DT), and logistic regression (LR).

4. Time complexity analysis for practical deployment.
 - Unlike previous studies that focus solely on accuracy, this research conducts a detailed analysis of training and testing times for each classifier.
 - This helps financial institutions choose models based on both performance and computational efficiency, making the approach more practical for real-world applications.
5. Development of a web-based loan approval prediction system.
 - A fully functional web application is developed using the Flask framework, enabling real-time loan approval predictions.
 - This ensures that the proposed model is not just theoretical but also deployable, bridging the gap between research and practical implementation in financial institutions.

The remainder of this paper is organized as follows. Section 2 reviews existing methods and techniques for loan approval. Section 3 describes the dataset used in this research. Section 4 outlines our proposed methodology for loan approval prediction. Section 5 presents the experimental results and discusses the performance of the proposed model. Section 6 discusses the findings in detail. Finally, Section 7 concludes the paper.

2. Literature Review

This section provides a comprehensive review of the relevant literature on the prediction of bank loan approvals using various ML and deep learning techniques. This review also identifies the limitations in previous studies listed in Table 1, highlighting areas where further research and improvement are required for more effective and accurate loan approval systems.

Omnia Elrashidy et al. (2024) [12] introduced an intelligent decision support system for loan evaluation using LR, DT, RF, SVM, K-nearest neighbors, and NB, incorporating data preprocessing and SMOTE. RF achieved the highest accuracy (81.9%), and a web application was developed for usability. Swati Srivastava (2024) [13] implemented LR, SVM, DT, RF, and an ensemble model, with the ensemble achieving 73.79% accuracy. Chen (2024) [14] trained eight models, highlighting AdaB with 84.95% accuracy and the highest F1-score, while XGB excelled in precision. Viswanatha et al. (2023) [9], Mahottam et al. (2023) [11], Nancy Deborah et al. (2023) [16], Hemanth Kumar Sarisa et al. (2023) [17], and Saini et al. (2024) [18] applied various combinations of RF, DT, NB, K-nearest neighbors, SVM, and support vector classifier. Across these studies, RF, NB, and SVC repeatedly achieved accuracies in the range of 81%–83.7%, demonstrating consistent reliability in loan prediction. Mahottam et al., in addition, used XAI (LIME) for model interpretability. Maurya et al. (2023) [15] and Divate et al. (2021) [20] applied SVM, outperforming DT and LR, highlighting the model's ability to minimize approval risks using features such as credit

score, income, and employment history. Kumar et al. (2022) [19] demonstrated the effectiveness of ensemble learning, combining DT with AdaB, which achieved superior accuracy over standalone classifiers including RF, SVM, and KNN. Doswalwar et al. (2021) [21], Sheikh et al. (2020) [22], Supriya et al. (2019) [24], Blessie et al. (2019) [25], and Vaidya (2017) [26] focused on traditional models such as LR, DT, NB, SVM, RF, and GB. These studies reported accuracies mostly between 78% and 81%, confirming the robustness of these classical algorithms for loan prediction. Alsaleem et al. (2020) [23] further included BayesNet and multilayer perceptron (MLP), with MLP reaching 80% accuracy, showing the potential of neural networks in loan risk prediction.

3. Dataset Construction

The dataset used in this research was collected from Kaggle, contributed by Debdata Chatterjee (owner) [27]. It pertains to loan approval prediction and includes 614 labeled records with 13 features, covering applicant demographics, loan details, and loan approval status. Kaggle is an online platform that hosts datasets and supports data science competitions. A detailed breakdown of the categorical and numeric attributes is provided in Table 2.

4. Methodology

This section outlines the detailed methodology employed in the study, including data preprocessing, feature selection using SHAP, classification using multiple ML models, and the integration of XAI techniques such as LIME to enhance the interpretability of the predictions. The overall approach aims to optimize predictive accuracy while ensuring transparency in model decision-making.

4.1. Data Preprocessing. It is a critical step to ensure the quality and reliability of the input data. This approach has allowed for uniform handling of missing values and categorical variables. The “Loan_ID” column was dropped as it was not relevant to the prediction task. Missing values were identified and handled using appropriate imputation methods. Categorical features such as “Gender,” “Married,” “Dependents,” “Self_Employed,” and “Credit_History” have been imputed using mode imputation, which has replaced missing values with the most frequent value in each column. Numerical features such as “LoanAmount” and “Loan_Amount_Term” have been imputed using an iterative imputer with a RF regressor. This advanced imputation technique has captured the underlying patterns in the data, resulting in more accurate and reliable imputations. After imputing missing values, categorical variables were encoded into numerical values. Binary features such as gender were mapped to integers, while other categorical features were transformed using one-hot encoding. Figure 1 shows the result of null value count and removal. The target variable exhibited moderate imbalance (approved $\approx$ 68%, not approved $\approx$ 32%). To prevent classifier bias toward the majority class, the training subset was balanced using the

TABLE 1: Summary of loan approval prediction studies using machine learning and deep learning algorithms.

Author(s) and year	Limitations
Omnia Elrashidy et al. (2024) [12]	1. While SMOTE was used for data balancing, it may introduce synthetic noise, impacting model reliability 2. The study does not apply or explore advanced feature selection methods to improve model performance and interpretability
Swati Srivastava (2024) [13]	1. Limited comparison to other advanced algorithms 2. Low accuracy
Chen (2024) [14]	1. Limited comparison to other advanced algorithms
Viswanatha et al. (2023) [9]	1. Focused only on accuracy, neglecting other metrics such as precision, recall, or AUC
Maurya et al. (2023) [15]	1. Limited comparison to other advanced algorithms
Mahottam et al. (2023) [11]	1. The study does not consider the time or computational resources required for training and prediction 2. The study does not apply or explore advanced feature selection methods to improve model performance and interpretability
Nancy Deborah et al. (2023) [16]	1. High testing time
Hemanth Kumar Sarisa et al. (2023) [17]	1. Limited comparison to other advanced algorithms 2. The study does not apply or explore advanced feature selection methods to improve model performance and interpretability
Saini et al. (2023) [18]	1. Limited comparison
Kumar et al. (2022) [19]	1. Limited comparison to other advanced algorithms
Divate et al. (2021) [20]	1. Focuses heavily on SVM, without exploring improvements through ensemble methods 2. Used a single splitting ratio, limiting model evaluation robustness 3. Low accuracy
Dosalwar et al. (2021) [21]	1. Limited comparison to other advanced algorithms 2. Low accuracy
Sheikh et al. (2020) [22]	1. Limited to logistic regression, missing the ability to capture nonlinear patterns
Alsaleem et al. (2020) [23]	1. Limited comparison to other advanced algorithms 2. Low accuracy
Supriya et al. (2019) [24]	1. No focus on preventing overfitting, especially with decision tree models
Blessie et al. (2019) [25]	1. Limited comparison to other advanced algorithms
Vaidya (2017) [26]	1. The study does not apply or explore advanced feature selection methods to improve model performance and interpretability 2. Low accuracy

TABLE 2: Summary of the dataset [27].

Feature name	Feature description	Data type
Loan_ID	Unique identifier for each loan	Object
Gender	Gender of the applicant (male/female)	Object
Married	Marital status of the applicant (Y/N)	Object
Dependents	Number of dependents the applicant has	Object
Education	Education level of the applicant (graduate/undergraduate)	Object
Self_Employed	Whether the applicant is self-employed (Y/N)	Object
ApplicantIncome	Income of the applicant	Integer
CoapplicantIncome	Income of the coapplicant	Integer
LoanAmount	Amount of loan in thousands	Float
Loan_Amount_Term	Term of the loan in months	Float
Credit_History	Whether the applicant has a good credit history	Float
Property_Area	The area type where the property is located (urban/semiurban/rural)	Object
Loan_Status	Loan approval status (Y/N)	Object

SMOTE algorithm. Numeric attributes such as ApplicantIncome and LoanAmount showed right-skewed distributions; hence, a logarithmic transformation was applied before model training.

4.2. Feature Selection Using SHAP. Feature selection plays a vital role in predictive modeling by improving model performance, reducing overfitting, and enhancing interpretability [28]. In this study, SHAP is employed to

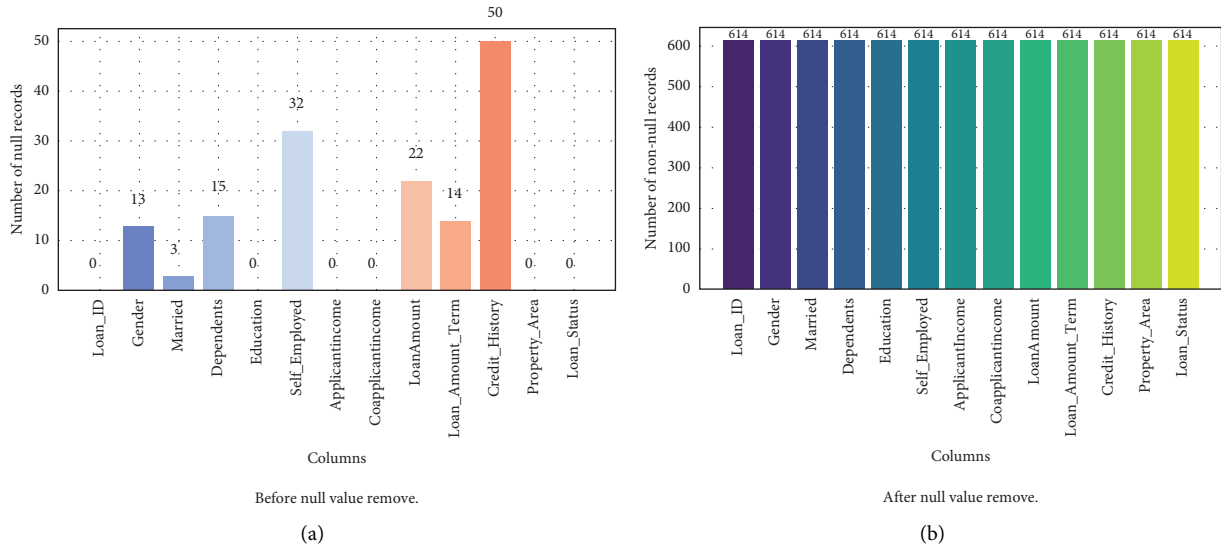


FIGURE 1: Before and after null value. (a) Before removing the null value. (b) After removing the null value.

quantify the contribution of each feature to the model's predictions. Grounded in cooperative game theory, SHAP provides consistent and locally accurate attributions, making it highly suitable for understanding both global and instance-level model behavior. Using SHAP importance scores, six key features, "Credit_History, Total_Income, LoanAmount, Property_Area, Education, and Dependents," were identified as the most influential predictors. Selecting these top-ranked features helps reduce dimensionality while ensuring that the model retains the variables most relevant to loan approval prediction. Figure 2 presents the ranked features based on their mean SHAP values, with higher values indicating greater influence on the prediction outcome.

To generate SHAP values, an RF classifier was trained on the full dataset, which includes variables such as "Gender, Married, Dependents, Education, Self-Employed, LoanAmount, Loan_Amount_Term, Credit_History, Property_Area, and Total_Income." RF was chosen due to its robustness, ability to model complex feature interactions, and strong generalization capability. The SHAP analysis derived from this model ensures that feature selection is both data-driven and theoretically grounded.

4.3. Building a Sophisticated Multiclassifier Framework.

To build a robust predictive model, several ML classifiers have been evaluated using the top six SHAP-selected features. Each classifier has been trained on the reduced feature set, and its performance has been assessed using various evaluation metrics. The multiclassifier framework is designed not only to benchmark accuracy but also to assess the generalizability of SHAP-selected features across diverse learning paradigms (e.g., linear models vs. ensemble methods). This provides empirical validation that the SHAP-based selection identifies domain-relevant predictors that remain stable across model families. The multiclassifier

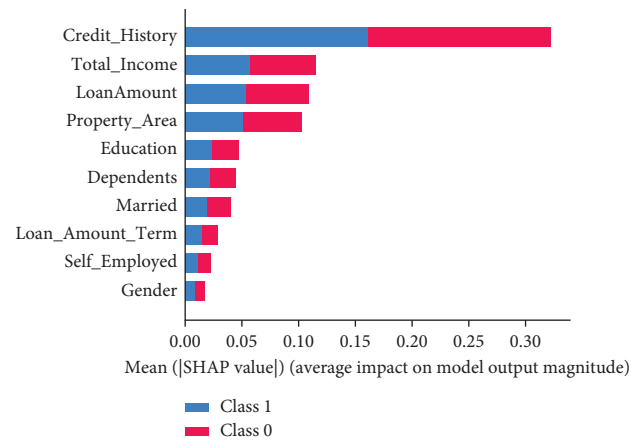


FIGURE 2: Features rank based on SHAP values for the dataset [27].

framework in this research consists of eight ML classifiers as follows:

- LR: A probabilistic model used for binary classification [29].
- RF: An ensemble-based model that aggregates multiple DTs to enhance accuracy and robustness [30].
- AdaB: A boosting algorithm that combines weak learners to create a strong model [31].
- XGB: An optimized GB algorithm known for its high efficiency and performance [32].
- GB: A sequential ensemble method that builds models iteratively to minimize prediction errors [33].
- NB: A probabilistic classifier based on Bayes' theorem, effective for categorical features [34].
- SVM: A classifier that finds the optimal hyperplane for data separation [35].
- DT: A rule-based classifier that recursively splits the data based on feature importance [36].

4.4. XAI Techniques Using LIME. To further enhance the transparency and interpretability of the ML models, XAI techniques have been incorporated. Specifically, LIME has been applied to explain individual predictions. LIME is a model-agnostic approach that explains the predictions of any classifier by approximating the model locally around the prediction of interest.

- **LIME explainer initialization:** A Lime-TabularExplainer has been initialized using the training data that contained the top six features. The explainer has been designed to work in a classification setting, generating local explanations for individual predictions. The explainer has been provided with the feature names and class labels (“not approved” and “approved”) for interpretability.
- **LIME explanation generation:** For each test instance, LIME has been used to generate explanations for the model’s prediction. These explanations have been in the form of feature contributions (with associated weights) that indicated how each feature influenced the prediction. For instance, LIME could show that a high Credit_History and low LoanAmount have contributed positively to a loan approval decision.

The abovementioned steps are illustrated in Figure 3.

5. Result Analysis

The evaluation of an ML model is conducted using specific performance metrics. The “scikit-learn” library’s matrix module provides essential functions for calculating various performance metrics. One of the key evaluation tools is the confusion matrix, which generates four results based on the dataset: true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). These outcomes are crucial for deriving important metrics such as accuracy, precision, TP rate (TPR), FP rate (FPR), TN rate (TNR), FN rate (FNR), and the F1-score, using the following formulas:

$$\text{accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{TP} + \text{FN} + \text{FP}}, \quad (1)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (2)$$

$$\text{sensitivity/TPR/recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (3)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad (4)$$

$$\text{specificity/TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \quad (5)$$

$$\text{FNR} = 1 - \text{TPR}, \quad (6)$$

$$\text{F1-score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}. \quad (7)$$

Figure 4 illustrates the performance of various ML classifiers in predicting bank loan approvals using an 80:20 training and testing split ratio. Among the eight classifiers evaluated, LR demonstrated the highest accuracy at 86.18%, outperforming all other models. NB followed closely with an accuracy of 85.37%, showcasing its effectiveness in handling probabilistic relationships within the dataset. RF also performed well, achieving an accuracy of 84.55%, reflecting its ability to capture nonlinear patterns by aggregating multiple DTs. AdaB and GB achieved accuracies of 81.3% and 82.93%, respectively, indicating their proficiency in enhancing weak learners to make better predictions. However, XGB, despite being known for its efficiency and accuracy in large datasets, scored a slightly lower accuracy of 78.86%. SVM and DT exhibited lower performance, with accuracies of 77.24% and 70.73%, respectively, signaling their limited suitability for this particular dataset in comparison to other ensemble methods. Overall, LR and NB emerged as the most reliable models for predicting loan approvals, while ensemble methods such as RF and GB offered competitive alternatives. The relatively lower performance of SVM and DT suggests that simpler models may struggle with the complexity and interdependence of financial and demographic features in this dataset.

The confusion matrix results in Table 3 present a detailed evaluation of various classifiers used in the bank loan approval prediction research, focusing on key metrics such as TPs, FNs, TNs, FPs, precision, recall, and F1-score. The SVM achieved perfect recall (1.00), correctly classifying all positive loan approvals, but with a lower precision (0.77) due to a significant number of FPs (28), which resulted in an F1-score of 0.87. LR and NB emerged as top performers with strong overall metrics. LR had a high recall (0.97) and precision (0.86), leading to an F1-score of 0.91. Similarly, NB showed excellent recall (0.98) and an F1-score of 0.91, although its precision was slightly lower (0.84). RF and GB both performed well, with RF achieving a precision of 0.88, a recall of 0.92, and an F1-score of 0.90, while GB had a slightly higher recall (0.93) and an F1-score of 0.89. XGB also demonstrated competitive performance, with a balanced precision (0.85) and recall (0.87), resulting in an F1-score of 0.86. AdaB, although similar to the other boosting methods, showed slightly lower precision (0.84) but maintained a high recall (0.92), achieving an F1-score of 0.88. The DT model, however, underperformed compared to the other classifiers, with a recall of 0.74 and an F1-score of 0.79, indicating difficulty in correctly identifying positive loan approvals. Although its precision (0.85) was comparable to that of other models, its higher FNs reduced its overall effectiveness. In conclusion, LR, NB, and RF proved to be the most reliable classifiers for this task, offering a good balance between precision and recall, while SVM’s perfect recall came at the cost of lower precision.

The performance evaluation in Table 4 provides insight into the TPR, FNR, TNR, and FPR for various classifiers used in the bank loan approval prediction research. The SVM achieved a perfect TPR of 1.00, meaning it correctly classified all positive cases, but this came at the cost of a 0.00 TNR and a high FPR of 1.00, indicating that it incorrectly

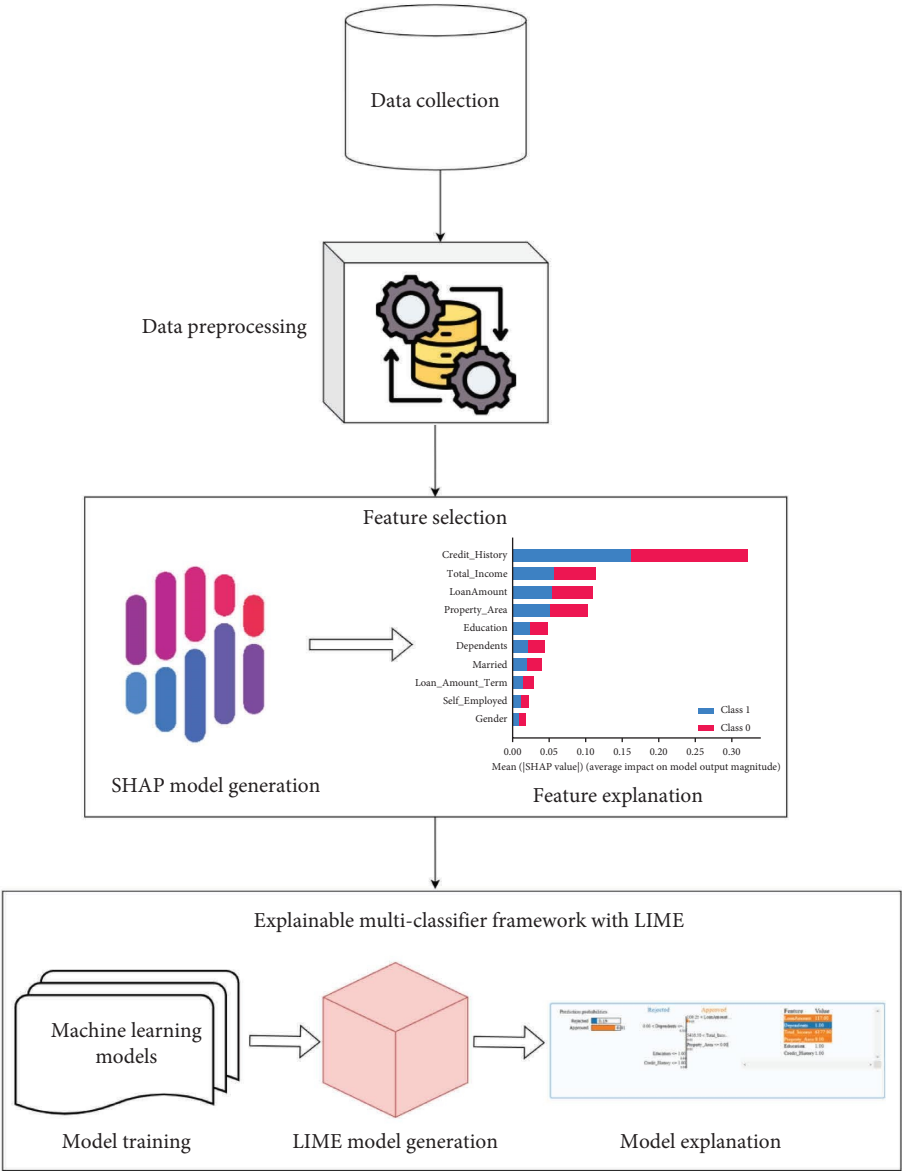


FIGURE 3: Block diagram of proposed methodology.

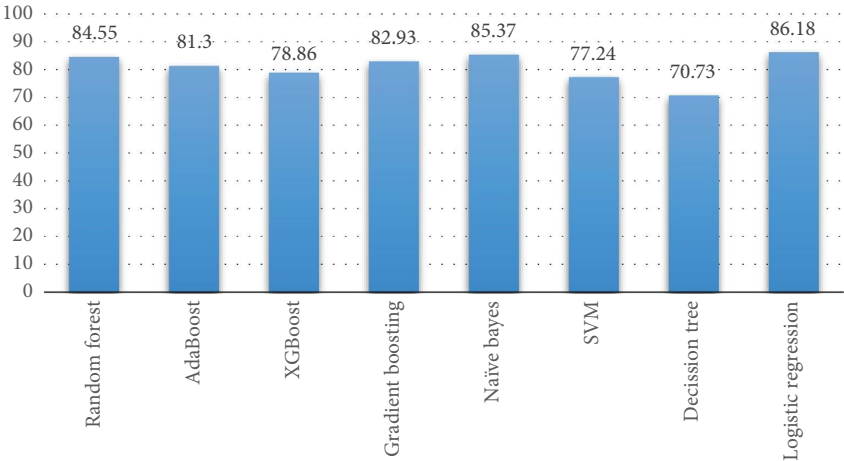


FIGURE 4: Classification accuracy comparison on dataset [27] using 80:20 split.

TABLE 3: Confusion matrix outcomes of different classifiers for the dataset [27].

Classifiers	TP	FN	TN	FP	Precision	Recall	F1-score
Random forest	88	7	16	12	0.88	0.92	0.90
AdaBoost	88	7	12	16	0.84	0.92	0.88
XGBoost	83	12	14	14	0.85	0.87	0.86
Gradient boosting	89	6	13	15	0.85	0.93	0.89
Naïve Bayes	94	1	11	17	0.84	0.98	0.91
Support vector machine	95	0	0	28	0.77	1.00	0.87
Decision tree	71	24	16	12	0.85	0.74	0.79
Logistic regression	93	2	13	15	0.86	0.97	0.91

TABLE 4: Performance evaluation matrices of different classifiers for the dataset [27].

Classifiers	TPR	FNR	TNR	FPR
Random forest	0.92	0.07	0.57	0.42
AdaBoost	0.92	0.07	0.42	0.57
XGBoost	0.87	0.12	0.50	0.50
Gradient boosting	0.93	0.06	0.46	0.53
Naïve Bayes	0.98	0.01	0.39	0.60
Support vector machine	1.00	0.00	0.00	1.00
Decision tree	0.74	0.25	0.57	0.42
Logistic regression	0.97	0.02	0.46	0.53

classified all negative cases as positive. This imbalance highlights the SVM's tendency to favor recall over precision. LR and NB both performed well in terms of TPR, with scores of 0.97 and 0.98, respectively, and very low FNRs of 0.02 and 0.01. However, both models had moderate TNRs (0.46 and 0.39, respectively) and elevated FPRs (0.53 and 0.60), indicating that while these models are strong at identifying positive instances, they tend to misclassify negative ones. RF also demonstrated a strong TPR of 0.92 and a relatively low FNR of 0.07, with a slightly better TNR of 0.57 compared to NB and LR, resulting in a lower FPR of 0.42. GB had a similarly high TPR of 0.93, an FNR of 0.06, and a moderate TNR of 0.46, yielding an FPR of 0.53. XGB, while still competitive, showed a slightly lower TPR of 0.87 and TNR of 0.50, with equal FPR and FNR values of 0.50 and 0.12, respectively. AdaB followed a similar trend, with a TPR of 0.92 but a lower TNR of 0.42 and a higher FPR of 0.57. The DT model, although it performed reasonably well with a TPR of 0.74, had the highest FNR (0.25) and a moderate TNR of 0.57, resulting in an FPR of 0.42. This suggests that the DT struggled more with identifying positive instances compared to the other classifiers. Overall, while SVM excelled in recall, RF, LR, and NB provided more balanced performances in terms of identifying both positive and negative instances, with RF showing the best trade-off between TPR and TNR, maintaining a lower FPR compared to others.

Figure 5 illustrates the AUC-ROC values for various ML classifiers applied to bank loan approval prediction, providing insight into their performance in distinguishing between approved and rejected loans. The AUC-ROC curve

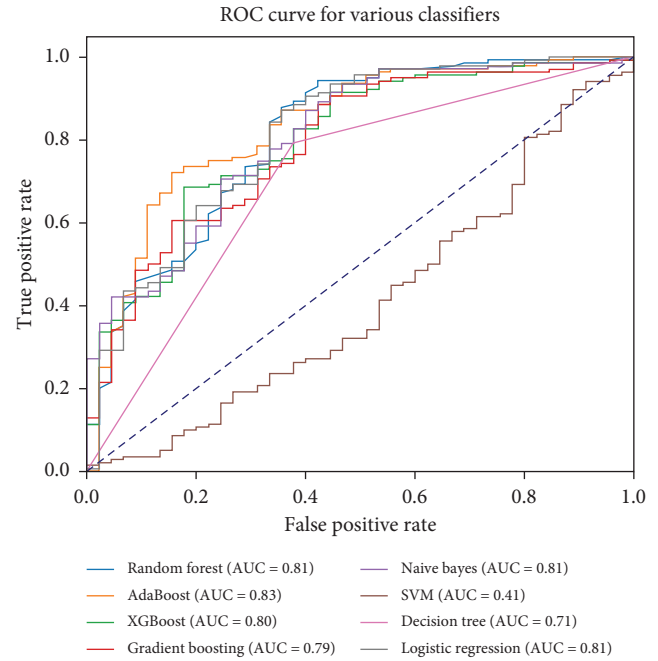


FIGURE 5: AUC values of different classifiers for dataset [27] using an 80:20 split.

is a crucial metric, as it reflects a model's ability to balance sensitivity and specificity. Among the classifiers, AdaB achieved the highest AUC-ROC score of 0.83, indicating it performed best in distinguishing between positive and negative classes. RF, NB, and LR all had strong AUC-ROC values of 0.81, reflecting their robustness and balanced performance in the prediction task. XGB, with an AUC-ROC of 0.80, performed slightly lower but remained competitive, while GB, scoring 0.79, was also effective but exhibited a marginally reduced ability to differentiate between classes. The DT model had a lower AUC-ROC of 0.71, suggesting that its predictive power was comparatively weaker in this task. Notably, the SVM displayed the poorest performance, with an AUC-ROC of just 0.41, indicating significant difficulties in accurately distinguishing between loan approvals and rejections. Overall, AdaB and the ensemble methods such as RF and NB showed strong performance, with LR also demonstrating its reliability in this classification task. However, SVM's low AUC-ROC suggests it may not be well-suited for this dataset and problem.

Table 5 presents the time complexity analysis, highlighting the training and testing times for various classifiers used in the bank loan approval prediction. Among the models, RF had the longest training time (0.534706 s) and a moderate testing time (0.015341 s), indicating it requires more computational resources for training but is relatively efficient during prediction. In contrast, AdaB and XGB had shorter training times, with 0.134333 and 0.142024 s, respectively, and their testing times were among the quickest, making them efficient choices for both training and real-time prediction tasks. NB stood out for its minimal training time (0.003380 s), demonstrating its efficiency in terms of computational cost, along with a fast-testing time of

TABLE 5: Time complexity analysis of different classifiers for the dataset [27].

Classifiers	Training time (s)	Testing time (s)
Random forest	0.534706	0.015341
AdaBoost	0.134333	0.007964
XGBoost	0.142024	0.004155
Gradient boosting	0.154048	0.001902
Naive Bayes	0.003380	0.002066
Support vector machine	0.090721	0.007247
Decision tree	0.005348	0.001931
Logistic regression	0.029752	0.002815

0.002066 s. LR also performed well with a low training time (0.029752 s) and a relatively quick testing time (0.002815 s). GB had a slightly higher training time (0.154048 s) but compensated with the lowest testing time (0.001902 s). DT was one of the fastest in both training (0.005348 s) and testing (0.001931 s), making it a good choice for quick computations, although it may lack predictive power compared to more complex models. The SVM had moderate training (0.090721 s) and testing times (0.007247 s), offering a balanced trade-off between computational cost and performance. Overall, NB, DT, and LR excelled in terms of computational efficiency, while ensemble methods such as RF, AdaB, and XGB required more time.

To evaluate the impact of SHAP-guided feature selection on classification performance, we compared it against two widely used baseline methods, recursive feature elimination (RFE) and mutual information (MI). Each selection technique retained the same number of top features (six) to ensure fair comparison. The models were trained under identical conditions using the same 80:20 split ratio. As shown in Table 6 SHAP-based feature selection consistently produced higher accuracy across most classifiers, improving average performance by approximately 3%–5% compared to traditional methods.

The optimized hyperparameters listed in Table 7 were selected through an exhaustive grid-search process to ensure fairness across all classifiers. These configurations directly correspond to the results presented in Figure 4. The tuning process substantially improved each model's predictive stability and accuracy relative to default settings. For instance, increasing the number of estimators and constraining the tree depth in the RF (200 trees, $\text{max_depth}=10$) prevented overfitting while yielding smoother learning curves and an accuracy gain of approximately 2%. Similarly, setting $n_estimators=250$ and $\text{learning_rate}=0.1$ in XGB balanced bias and variance, producing a consistent improvement in both accuracy and recall. The AdaB and GB models achieved optimal convergence with moderate learning rates (0.8 and 0.1, respectively), confirming the benefit of careful regularization over aggressive boosting. LR achieved its highest accuracy (86.18%) using the LIBLINEAR solver with L2 regularization ($C=1.0$), demonstrating that even a simple linear model can outperform complex ensembles when tuned properly and guided by SHAP-selected features. The SVM with an RBF kernel ($C=1.0$, $\gamma=\text{"scale"}$) provided a stable boundary

TABLE 6: Comparison of model accuracies under different feature selection techniques.

Classifiers	RFE	MI	SHAP
Logistic regression	82.47	83.02	86.18
Random forest	81.36	82.04	84.55
AdaBoost	78.92	79.33	81.30
XGBoost	75.60	76.15	78.86
Gradient boosting	79.84	80.21	82.93
Naïve Bayes	83.42	84.15	85.37
Support vector machine	74.05	75.18	77.24
Decision tree	68.94	69.73	70.73

between classes, though its performance was slightly limited by overlapping feature regions in the dataset. NB and DT benefited marginally from tuning of the var_smoothing and max_depth parameters, respectively, improving generalization while maintaining low computational cost. Overall, the tuned configurations confirm that parameter optimization plays a crucial role in achieving the robust accuracies reported in Figure 4. LR, RF, and NB emerged as the most reliable classifiers after tuning, validating that the model selection and optimization pipeline adopted in this study yields reproducible, high-quality results suitable for real-time financial decision-support systems.

To enhance the interpretability of LR, XAI techniques, specifically the LIME method, were employed. LIME helps to explain individual predictions by approximating the behavior of the complex model with a simpler, interpretable model. Specifically, the LimeTabularExplainer was utilized to generate explanations for specific data points in the bank loan approval prediction dataset, highlighting how different features contribute to the model's decision. The LIME explainer was applied to a test record, and the corresponding explanation visualizes the impact of the top 6 most influential features used in the model. By illustrating how each feature influences the prediction, whether for loan approval or rejection, the model's decisions become more transparent and understandable. Figure 6 offers detailed LIME explanations, giving clear insights into the significance of each feature for a specific record.

To enhance experimental robustness, 5-fold and 10-fold cross-validation were applied to each classifier in Table 8. Cross-validation ensures that every sample is used for both training and testing, minimizing overfitting and improving the reliability of comparative results. The results demonstrate stable performance across folds, confirming that the models generalize well. LR consistently yields the highest accuracy under both validation schemes, showing that its predictive capability is not a consequence of random train-test splits. Ensemble methods such as RF and GB maintain strong performance with slightly higher variance, typical of tree-based learners. The small incremental improvement from 5-fold to 10-fold ($\approx 0.4\%$) indicates that the dataset is sufficiently large for 5-fold CV to provide reliable estimates while balancing computational cost.

To assess the model's efficiency, a comparison was conducted with the results of the models proposed by other authors, as presented in Table 9.

TABLE 7: Summary of tuned hyperparameters.

Classifier	Tuned parameters (best values)
Logistic regression	Solver = "LIBLINEAR"; C = 1.0; penalty = "l2"
Random forest	$n_estimators = 200$; max_depth = 10; min_samples_split = 4
AdaBoost	$n_estimators = 100$; learning_rate = 0.8
XGBoost	$n_estimators = 250$; learning_rate = 0.1; max_depth = 6
Gradient boosting	$n_estimators = 200$; learning_rate = 0.1; max_depth = 3
SVM	Kernel = "rbf"; C = 1.0; gamma = "scale"
Naïve Bayes	var_smoothing = $1e-9$
Decision tree	Criterion = "Gini"; max_depth = 8



FIGURE 6: LIME explanation for loan approval prediction (Record no. 15).

TABLE 8: Comparison of model accuracies using 5-fold and 10-fold cross-validation.

Classifier	5-fold CV	10-fold CV
Logistic regression	86.04	86.31
Random forest	84.25	84.78
AdaBoost	81.11	81.56
XGBoost	78.64	79.02
Gradient boosting	82.77	83.10
Naïve Bayes	85.15	85.46
Support vector machine	77.05	77.41
Decision tree	70.54	70.92

To evaluate the efficiency of the proposed model, an additional dataset obtained from Kaggle [37] was utilized. This dataset comprises 4269 instances, with 12 independent variables and one dependent variable. A detailed description of all attributes in this dataset is presented in Table 10.

The six most significant features (CIBIL_score, loan_term, loan_amount, income_annum, residential_assets_value, and luxury_assets_value) have been selected based on the SHAP importance scores. This approach reduces the dimensionality of the dataset, focusing only on the most relevant predictors. In Figure 7, features are ranked by their average SHAP values, with the most important features at the top and the least important ones at the bottom.

Table 11 presents the performance evaluation metrics of various classifiers applied to the dataset [27]. The evaluation is based on multiple criteria, including accuracy, TPs, FNs, TNs, FPs, precision, recall, and F1-score, which provide a comprehensive understanding of the classifiers'

TABLE 9: Comparison with state-of-the-art works using the same dataset [27].

Author(s) and year	Accuracy (%)
Omnia Elrashidy et al. (2024) [12]	81.9
Swati Srivastava (2024) [13]	73.79
Chen (2024) [14]	84.95
Viswanatha et al. (2023) [9]	83.74
Maurya et al. (2023) [15]	85.0
Mahottam et al. (2023) [11]	83.33
Nancy Deborah et al. (2023) [16]	83
Hemanth Kumar Sarisa et al. (2023) [17]	83.45
Saini et al. (2023) [18]	81.04
Kumar et al. (2022) [19]	84.0
Divate et al. (2021) [20]	65.83
Dosalwar et al. (2021) [21]	78.5
Sheikh et al. (2020) [22]	81.0
Alsalem et al. (2020) [23]	80.0
Supriya et al. (2019) [24]	81.0
Blessie et al. (2019) [25]	80.04
Vaidya (2017) [26]	65.52
Proposed model + LR	86.18

Note: The bold value indicates higher accuracy than the others.

effectiveness. Among the classifiers, LR achieved the highest accuracy (0.990632), closely followed by RF (0.989461) and GB (0.988290). These models demonstrate strong predictive power, maintaining high precision and recall values, indicating their ability to correctly classify both positive and negative cases with minimal errors. RF and LR exhibit superior F1-scores (0.991485 and 0.992424, respectively), suggesting a well-balanced trade-off between precision and recall. This implies their robustness in correctly identifying

TABLE 10: Dataset details [37].

Feature name	Feature description	Data type
laniid	Unique identifier for each loan application	int64
no_of_dependents	Number of dependents of the applicant	int64
Education	Educational qualification of the applicant	Object
self_employed	Employment status (self-employed or not)	Object
income_annum	Annual income of the applicant	int64
loan_amount	Loan amount requested by the applicant	int64
loan_term	Duration of the loan in months	int64
CIBIL_score	CIBIL credit score of the applicant	int64
residential_assets_value	Value of residential assets owned	int64
commercial_assets_value	Value of commercial assets owned	int64
luxury_assets_value	Value of luxury assets owned	int64
bank_asset_value	Value of bank assets owned	int64
loan_status	Final loan approval status	Object

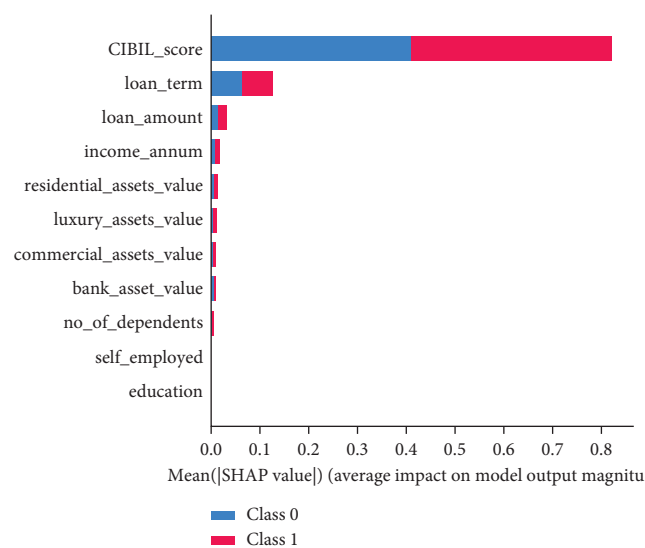


FIGURE 7: Features rank based on average SHAP values for the dataset [37].

both positive and negative instances while minimizing misclassification errors. AdaB and DT also demonstrate competitive performance, with accuracies of 0.967213 and 0.978923, respectively. These models maintain high precision and recall values, making them reliable choices for classification tasks. Conversely, XGB and NB exhibit lower accuracy scores (0.728337 and 0.761124, respectively), with relatively higher FPs and FNs. The significantly lower TNs for XGB (123) and NB (134) indicate their struggle in correctly identifying negative cases, affecting their overall performance. SVM achieves the lowest accuracy (0.615925), despite having a perfect recall value of 1.000000. The high FPR (328 FP) suggests that while SVM classifies all positive instances correctly, it misclassifies a substantial number of negative cases, leading to poor precision and overall lower effectiveness.

6. Discussion

This research presents an ML model for loan approval prediction, incorporating SHAP-based feature selection and

LIME for model interpretability within a multiclassifier framework. Although SHAP and LIME have individually been employed in prior research, this work uniquely combines them in a cohesive framework where SHAP is used not only for interpretability but also as a premodel feature selection mechanism. This dual role of SHAP enhances model efficiency while maintaining explainability. The study's contribution, therefore, lies in demonstrating that an interpretable and computationally efficient model, such as LR, can outperform complex ensemble methods when guided by XAI feature selection. The model aimed to enhance the accuracy of predictions while maintaining transparency in decision-making, an essential factor for real-world applications in financial institutions. The approach involved training eight distinct classifiers, with LR demonstrating the best performance, achieving an accuracy of 86.18% for dataset [27], followed closely by NB and RF, and 99.06% for dataset [37]. Although AdaB achieved the highest AUC (0.83), LR demonstrated the highest accuracy (86.18%) and maintained strong AUC performance (0.81). The slight difference arises from AdaB's bias toward recall under class imbalance, while LR achieves a better balance between precision and recall. Given that banking applications demand explainable, reliable, and easily auditable models, LR was considered the most appropriate classifier despite the marginally lower AUC. One of the significant contributions of this research is the integration of XAI techniques such as SHAP and LIME. SHAP helped to identify the most important features contributing to loan approval decisions, such as credit history and loan amount, reducing the dimensionality of the problem and making it easier to interpret the outcomes. LIME provided further insight by offering local explanations for individual predictions, which enhances the trustworthiness of the model for stakeholders.

The proposed loan approval prediction framework demonstrates several key strengths that distinguish it from prior studies. The model achieves high predictive accuracy, with LR reaching 86.18%, outperforming earlier approaches and providing reliable decision support. This performance is further enhanced by SHAP-based feature selection, which identifies the most influential predictors, ensuring the model focuses on the most relevant information. In addition,

TABLE 11: Performance evaluation matrices of different classifiers for the dataset [37].

Classifier	Accuracy	TP	FN	TN	FP	Precision	Recall	F1-score
Random forest	0.989461	524	2	321	7	0.986817	0.996198	0.991485
AdaBoost	0.967213	511	15	315	13	0.975191	0.971483	0.973333
XGBoost	0.728337	499	27	123	205	0.708807	0.948669	0.811382
Gradient boosting	0.988290	523	3	321	7	0.986792	0.994297	0.990530
Naive Bayes	0.761124	516	10	134	194	0.726761	0.980989	0.834951
SVM	0.615925	526	0	0	328	0.615925	1.000000	0.762319
Decision tree	0.978923	516	10	320	8	0.984733	0.980989	0.982857
Logistic regression	0.990632	524	2	322	6	0.988679	0.996198	0.992424

LIME-based interpretability provides transparency at the individual prediction level, fostering trust and explainability in financial decisions. The framework also benefits from computational efficiency through the use of LR and NB, enabling faster training and prediction suitable for real-time applications. Beyond performance, the multiclassifier architecture offers flexibility and scalability, allowing adaptation to diverse datasets and smooth deployment within banking systems.

Despite these strengths, several limitations remain. The datasets used originate from Kaggle and may not fully represent real banking environments with evolving credit policies. Furthermore, fairness and bias assessment are not addressed, which is critical for ensuring equitable financial decisions. The current framework also lacks mechanisms for temporal drift monitoring and feature stability under changing socioeconomic conditions. While SHAP and LIME enhance interpretability, practical deployment would require integrating model auditing tools and human-in-the-loop oversight.

7. Conclusion and Future Work

This research presents a comprehensive approach to enhancing loan approval predictions using ML techniques, focusing on both accuracy and interpretability. The model integrates SHAP-guided feature selection to identify the most influential predictors, ensuring the model remains both efficient and explainable. In addition, the use of LIME for individual prediction explanations adds an essential layer of transparency, making the model's decisions clearer to stakeholders, which is particularly important in sensitive areas such as financial decision-making. Among the eight classifiers used, LR demonstrated the highest accuracy, achieving 86.18%, which is competitive with state-of-the-art models. This high performance highlights the model's reliability in predicting loan approval outcomes, offering banks and financial institutions a robust tool for risk management and decision-making. Overall, the proposed SHAP-guided and LIME-enhanced multiclassifier framework demonstrates that explainable, transparent, and highly accurate loan approval prediction systems can be achieved using interpretable models such as LR. The findings have practical implications for financial institutions by enabling fair, auditable, and data-driven loan decision processes. Future work will focus on addressing current limitations by incorporating real-world bank datasets, fairness-aware learning, drift monitoring, hybrid deep-XAI models, and

automated model governance modules to ensure robust, ethical, and adaptive deployment, thereby strengthening reliability, generalizability, and regulatory compliance.

Data Availability Statement

All datasets and code are available at the following links:

1. Datasets and source code: https://github.com/rajib1346/Bank_Loan-Source_Code.git.
2. Web app source code: https://github.com/rajib1346/Bank_Loan_App_EXAI.git.

Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

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